Module 4 Lab 2: First boto3 Calls and Token Cost Math

20 points. Estimated time 100 to 130 minutes.

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How to use this file. Type your answers straight into the shaded boxes. Do not delete the questions. Anything marked as written in advance has to be filled in before you run that step, not afterwards. When you are finished, save this file and upload it with the rest of your submission.

1. Your unique values

These are what make the submission yours. They come from your own account and your own run, so no two students have the same ones.

Your Region: us-east-1

The model you chose and why: deepseek.v3.2. It showed up in my model list and my account could call it right away.

Your measured input and output token counts:
Prose: 39 in, 170 out. Structured: 82 in, 46 out.

Your cost per call and the three volume figures:
$0.00012484 per call. 1,000/mo = $0.12, 50,000/mo = $6.24, 2,000,000/mo = $249.68

2. Predictions, written in advance

Write each prediction BEFORE you run that step. A wrong prediction reasoned honestly earns full credit. One edited afterwards to match the result earns none.

Step 2

Before you run it, predict: will the number of models VISIBLE in your Region equal the number you have ACCESS to?

Yes

What actually happened

99 models listed in us-east-1. deepseek.v3.2 worked on the first call, no access errors.

Why it was different, or why I was right

All available models are listed

Step 4

Before you run the second set, predict: at temperature 1.0, will the five answers differ from each other more, less, or the same as at temperature 0.2? And will output token counts vary more?

More, but counts should be similar.

What actually happened

The texts differed more but token counts were similar

Why it was different, or why I was right

Temperature affects token variability, not output length

3. Break it on purpose

Working code shows you what happens. Broken code shows you why. Do each of these deliberately, paste the real error text, and undo the change before moving on.

Break 1. Call converse with a modelId you do not have access to.

Paste the exact error text here, not a description of it

ValidationException: An error occurred (ValidationException) when calling the Converse operation: The provided model identifier is invalid.

The exact error text. Note whether it tells you the model does not exist or that you lack access, because those are different problems.

ValidationException: An error occurred (ValidationException) when calling the Converse operation: The provided model identifier is invalid.
It says the identifier is invalid

Break 2. Set maxTokens to 5 and re run your prose prompt.

Paste the exact error text here, not a description of it

'stopReason': 'max_tokens', 'usage': {'inputTokens': 39, 'outputTokens': 5, 'totalTokens': 44}, 'metrics': {'latencyMs': 4198}}

What comes back and what stopReason says. Note that you are billed for what was generated regardless.

'stopReason': 'max_tokens', 'usage': {'inputTokens': 39, 'outputTokens': 5, 'totalTokens': 44}, 'metrics': {'latencyMs': 4198}}

Stop reason says 'max_tokens'

Break 3. Use the control plane client to call converse, and the runtime client to call list_foundation_models.

Paste the exact error text here, not a description of it

AttributeError: 'Bedrock' object has no attribute 'converse'
AttributeError: 'BedrockRuntime' object has no attribute 'list_foundation_models'

Both exact errors. This is the mistake from the first mini-tutorial, and seeing it once means you will recognize it instantly later.

AttributeError: 'Bedrock' object has no attribute 'converse'
AttributeError: 'BedrockRuntime' object has no attribute 'list_foundation_models'

4. Results

The numbers and outputs your own run produced.

Your Region and chosen model, with the reason

us-east-1, deepseek.v3.2. It showed up in my model list and my account could call it right away.

Your measured token counts for both prompt designs

Prose: 39 in / 170 out (209 total)
Structured: 82 in / 46 out (128 total)
Output ratio: 170 / 46 = about 3.7x

Your latency and variability measurements at both temperatures

Temp 1.0 = 'totalTokens': 196
Temp 0.2 = 'totalTokens': 179

Your cost table with source and date

Rates: $0.58 per 1M input, $1.68 per 1M output (aws.amazon.com/bedrock/pricing, read 11/11/2026)
Per call: (82/1000 x 0.00058) + (46/1000 x 0.00168) = $0.00012484
1,000/mo = $0.12, 50,000/mo = $6.24, 2,000,000/mo = $249.68

5. Reflection

Your own words. No AI assistance on this part. Graders check these answers against the rest of your submission.

Question 1

Report your Step 2 prediction about visible models versus accessible models, and the two actual numbers from your account. Say what you had to do to close the gap, if anything.

I predicted the would be the same. 99 models were listed and accessible. All models are now automatically accessible

Question 2

Report your Step 4 prediction about temperature and what your ten measurements actually showed. Quote at least two of your own latency figures and two output token counts.

I predicted they would be the same and they showed the same token count but latency differed

Temp 1.0 = {'totalTokens': 196}, 'metrics': {'latencyMs': 5208}
Temp 0.2 = {'totalTokens': 179}, 'metrics': {'latencyMs': 5677}

Temp 1 count: 160.6 Temp 1 range: 103 - 200
Temp 0.2 count: 150.777 Temp 0.2 range: 129 - 200

Temp 1 Latency: 4412.6 Temp 1 Latency range: 2483 - 6098
Temp 0.2 Latency: 5146.3 Temp 0.2 Latency range: 3983 - 8420

Question 3

Quote your own token counts for the prose prompt and the structured prompt from Step 5. Compute the ratio and say what it implies for how you will write prompts for the rest of this course.

prose input 39 output 170
structured input 82 output 46
output token ratio (prose / structured) = 3.7x

I will write structured prompts

Question 4

Your cost table shows a figure at two million calls per month. Say whether you would approve that feature if it were your budget, and name the specific thing you would want measured before deciding.

I would approve that feature if it could be showed that the cost was less than the cost of existing methods or a person doing it

Question 5

In your third break it exercise you swapped the two clients. Explain in your own words what the control plane and the runtime plane each do, as though telling a classmate who is about to make the same mistake.

Runtime runs the service, control plane configures it

6. AI use disclosure

Required by the course AI Use Policy.

Tool used (write none if you used no AI):
Claude (Claude Code)

If you used one: what you asked it, what you kept, and how you checked it

Asked it to build a notebook skeleton from the lab page
Kept the code layout. I picked the model, prompts and rates and ran everything on my own account.
Checked the numbers here against my notebook output.

7. Before you submit

- ☐ I printed the full response dictionary at least once
- ☐ My cost table uses my own measured token counts, not the example ones
- ☐ I captured the exact error for a model I do not have access to
- ☐ Every prediction was written before I ran that step
- ☐ I recorded the actual result for every prediction, including the wrong ones
- ☐ I pasted the real error text for every break it item
- ☐ My notebook runs top to bottom with every output visible
- ☐ PDF exported after that full run, and it matches
- ☐ Work pushed to my GitHub portfolio repository and the URL is in here
- ☐ Reflection written by me with no AI assistance
- ☐ Files uploaded individually, no archive